

Bridget Crampton

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EDUCATION

Northeastern University, Khoury College of Computer Sciences

Boston, MA

Candidate for Bachelor of Science in Data Science – GPA: 3.732/4.0

2028

Relevant Coursework: Data Structures, Algorithms, Machine Learning, Statistics, Data Science, Advanced Programming with Data, Linear Algebra, Calculus, Probability Theory, Databases Systems, Data Visualization, Financial Accounting and Reporting

TECHNICAL SKILLS

Languages: Python, SQL

ML/Data Science: Scikit-Learn, XGBoost, TensorFlow/Keras, NumPy, Pandas, SciPy, NLTK, NetworkX

Databases & Tools: PostgreSQL, Snowflake, dbt, PySpark, Beautiful Soup, Selenium, Git/Github, Jupyter Notebook

Visualization: Tableau, Matplotlib, Seaborn, Plotly, Altair, D3.js

PROFESSIONAL EXPERIENCE

Incoming Business Intelligence Co-op | Klaviyo, Boston, MA

July 2026 – December 2026

- Selected for the Business Intelligence team to support data-driven decision making and analytics initiatives across the organization

Data Analyst Intern | IDX Exchange, Remote

March 2026 – June 2026

- Constructed **Python/Pandas** pipeline ingesting 12+ months of MLS transaction data via **CoreLogic Trestle API**, engineering 7+ market metrics (price-to-list ratio, price per sq ft, listing-to-contract days) with **IQR** outlier detection
- Developed 6 interactive **Tableau** dashboards covering market trends and top-100 agent/brokerage competitive rankings with zip code heat maps, filterable by geography and property type; presented findings in a final market intelligence report

Undergraduate Research Assistant | Network Science Institute, Boston, MA

Nov 2025 – April 2026

- Engineered **Python** pipeline transforming **StatsBomb** event data into possession networks with **NetworkX**, computing graph metrics (betweenness centrality, PageRank, density) and spatial features using **Delaunay/Voronoi** to analyze team coordination
- Designed tactical pattern detection system classifying 7 attack types using **k-means clustering** on possession signatures, comparing goal-scoring vs. non-goal sequences to identify high-value passing networks and spatial configurations

Data Science Intern | SparkJar Agency, Seattle, WA

May 2025 – July 2025

- Built **Python** web scraper using **Beautiful Soup** and **Regex** to extract and structure SEC filing data from 30+ companies, performing data cleaning and transformation to support client BI workflows
- Developed NLP-based AI agent using **Google ADK** to analyze SEC text data and identify credit risk patterns for small businesses, delivering quantitative insights and actionable recommendations to stakeholders

PROJECTS

Credit Card Fraud Detection: Comparative ML Study | Academic Project – Machine Learning

Spring 2025

- Trained 6 models on 590k+ IEEE-CIS transactions (434 features), achieving **XGBoost** AUC of 0.9075 (0.9195 public Kaggle leaderboard); engineered 10+ features including card-level aggregations and amount deviation, improving AUC by 0.0012
- Implemented custom weighted binary cross-entropy loss, **XGBoost**-guided hybrid neural architecture using top-50 features, and stacking ensemble achieving 0.67 fraud precision vs. 0.26 for standalone XGBoost

Co-opScout | Personal Project

Nov 2025 – Jan 2026

- Developed **Python/Selenium** web scraper with cookie-based authentication to extract 1000+ NUworks co-op postings, implementing data validation, duplicate prevention, and **JSON** export with structured cleaning pipelines
- Engineered **Flask** REST API with **Supabase/PostgreSQL** integration and content-based recommendation system using **TF-IDF** vectorization and cosine similarity to match students with co-op opportunities

LEADERSHIP & CAMPUS INVOLVEMENT

Software Product Lab Member | Forge, Northeastern University, Boston, MA

Jan 2026 – May 2026

- Engineered **AST**-based parser using **Tree-sitter** to extract 10+ structural metrics per file (cyclomatic complexity, class hierarchies, dependencies), transforming unstructured codebases into structured graph data for 3D visualization

Internal Tools Engineer | Generate Product Development Studio, Northeastern University, Boston, MA

Jan 2026 – May 2026

- Built a full-stack application portal for Generate NU using **React**, **Vite**, **Clerk**, and **Supabase**, implementing role-based auth, configurable application forms, admin review tooling, and AI-assisted UI development

Founder, Co-President of Outreach | Girls Who Code Chapter, Northeastern University, London, UK

Jan 2025 – May 2025

- Grew chapter 5x from 4 to 50+ members; organized semester-long speaker series spanning agritech, data visualization, and space exploration for 80+ attendees